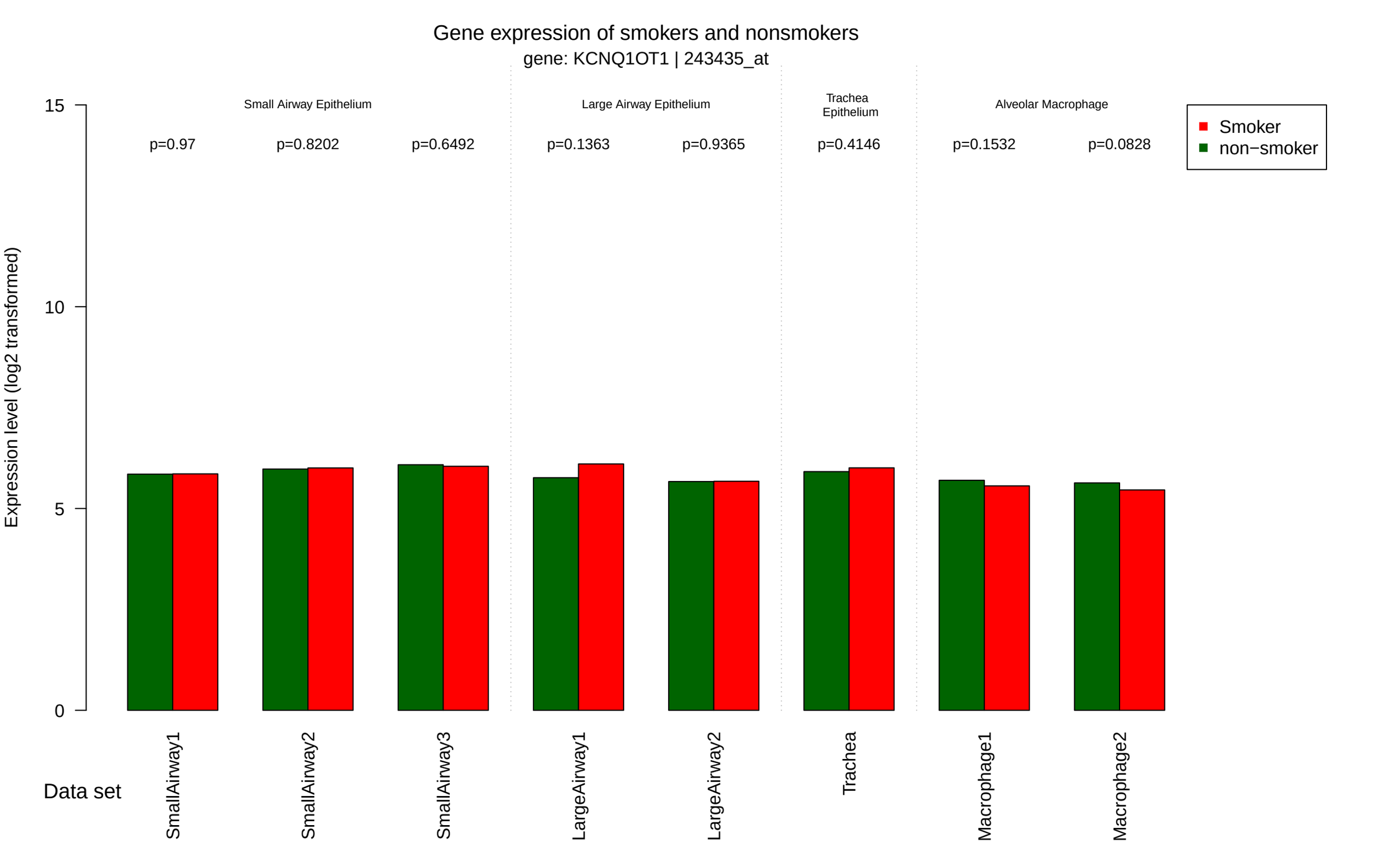
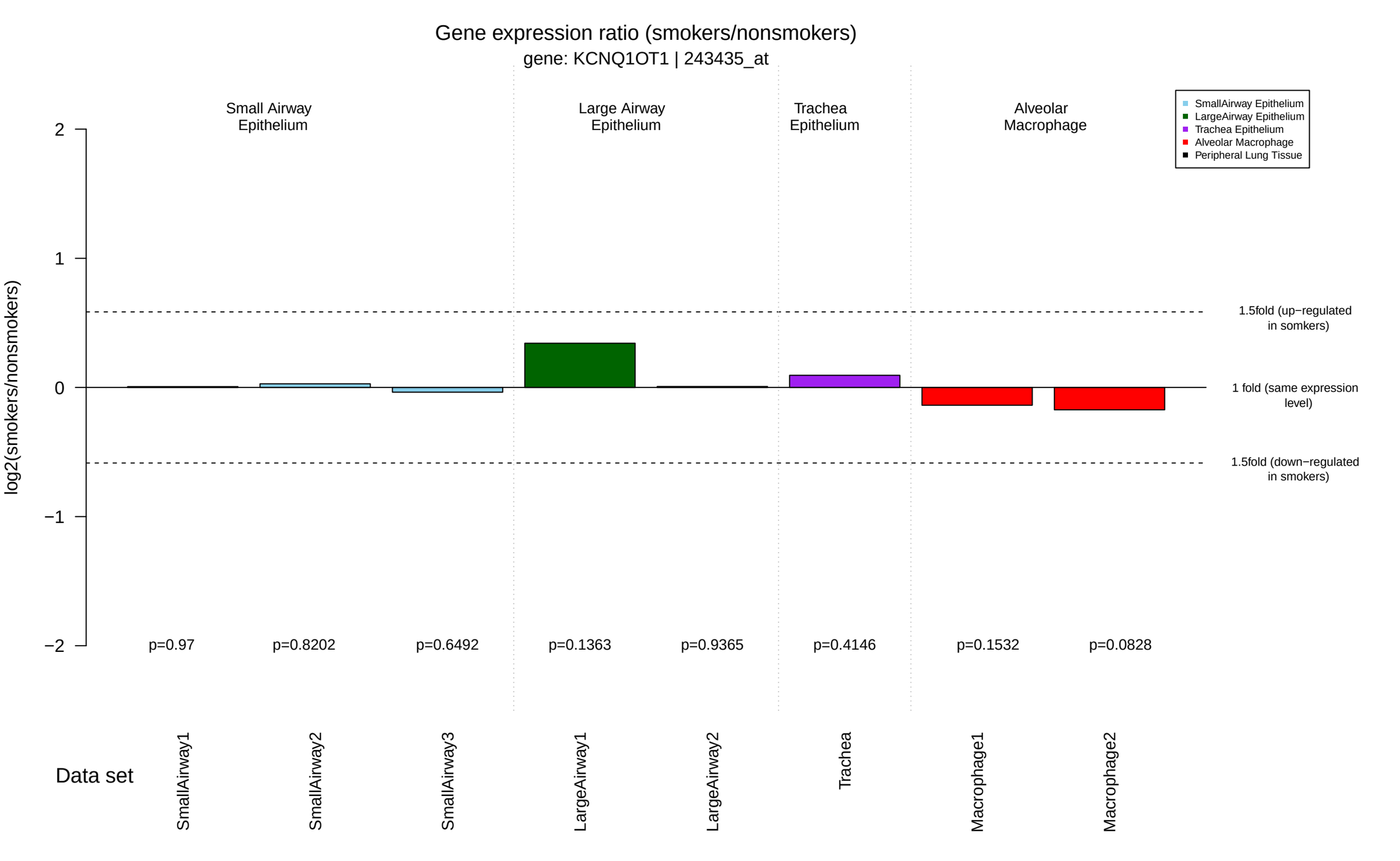
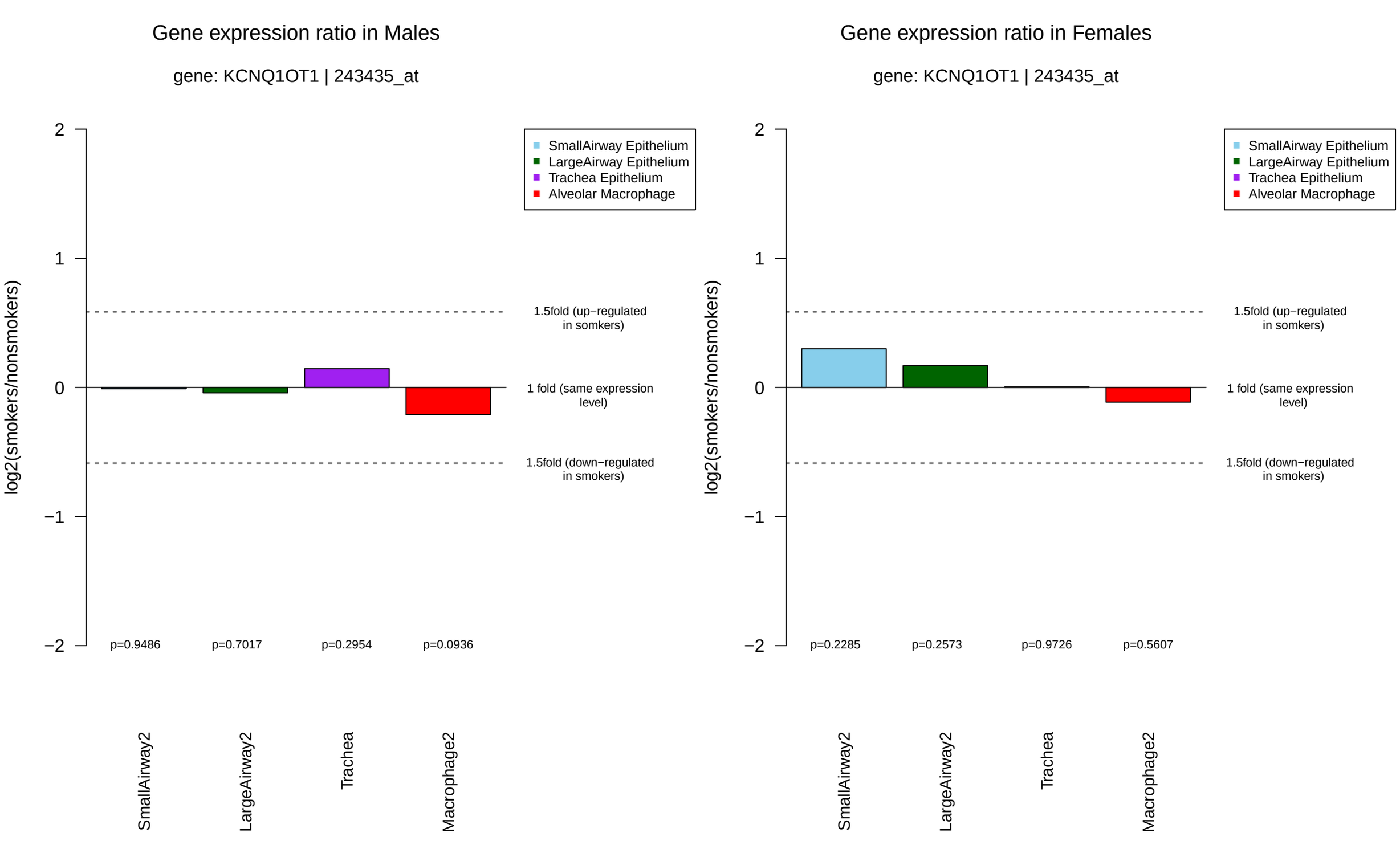
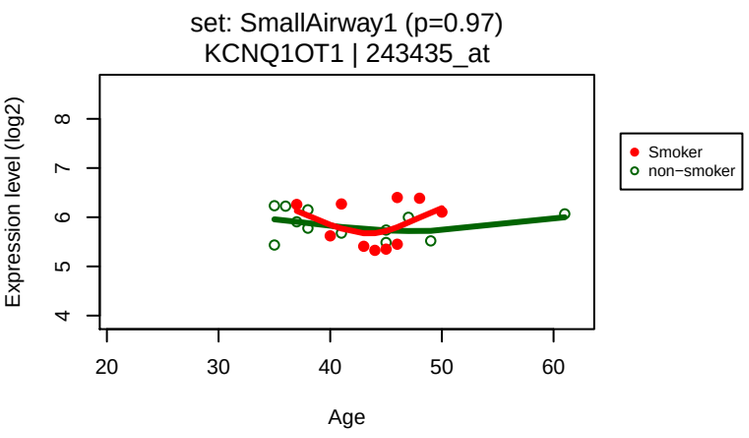
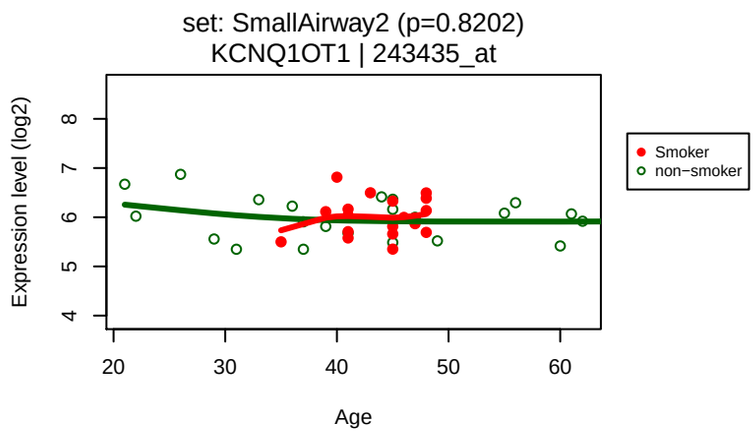
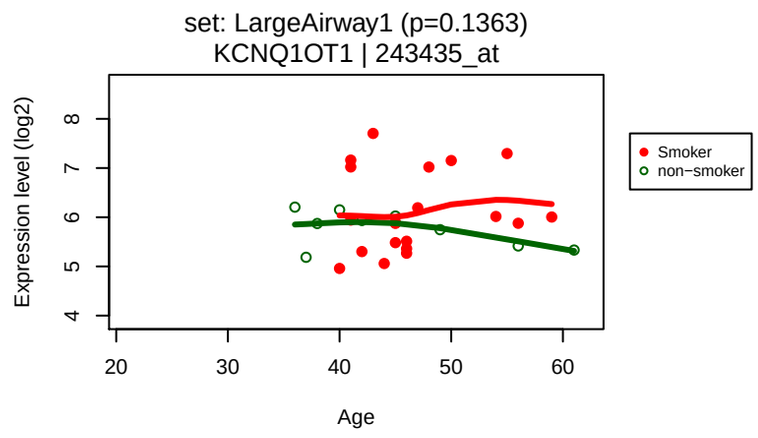

Gene expression of smokers and nonsmokers
gene: KCNQ1OT1 | 243435_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.97
p=0.8202
p=0.6492
p=0.1363
p=0.9365
p=0.4146
p=0.1532
p=0.0828
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

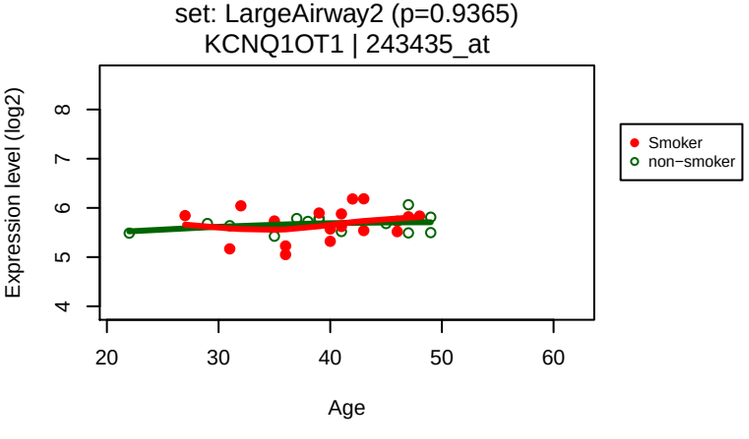

Gene expression ratio (smokers/nonsmokers)
gene: KCNQ1OT1 | 243435_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.97
p=0.8202
p=0.6492
p=0.1363
p=0.9365
p=0.4146
p=0.1532
p=0.0828
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

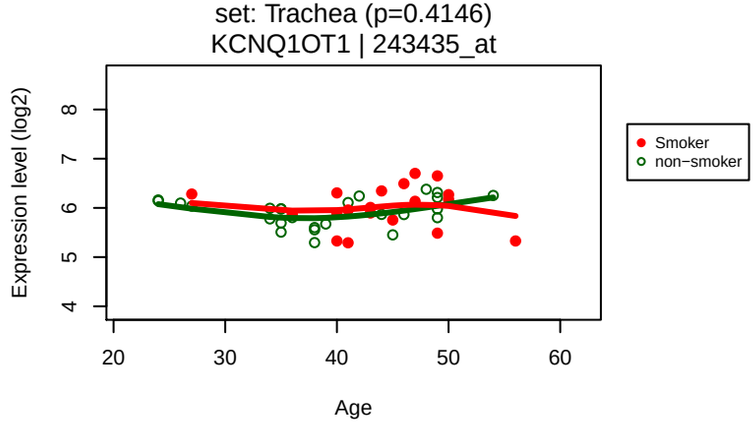

Gene expression ratio in Males
gene: KCNQ1OT1 | 243435_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9486
p=0.7017
p=0.2954
p=0.0936
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNQ1OT1 | 243435_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2285
p=0.2573
p=0.9726
p=0.5607
SmallAirway2
LargeAirway2
Trachea
Macrophage2

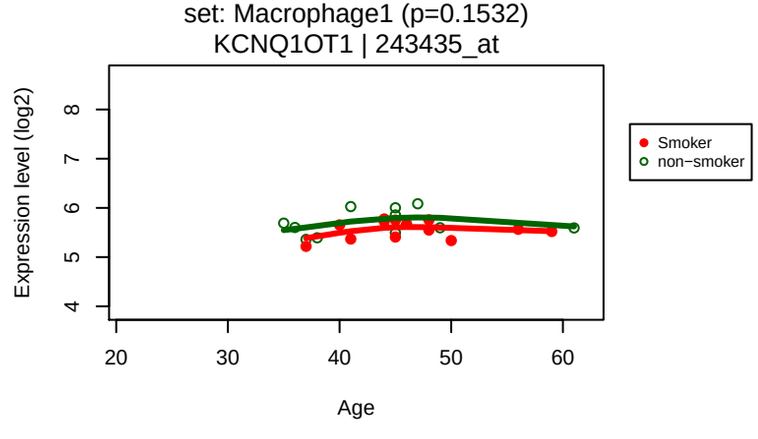

set: SmallAirway1 (p=0.97)
KCNQ1OT1 | 243435_at
Expression level (log2)
Age
Smoker
non−smoker

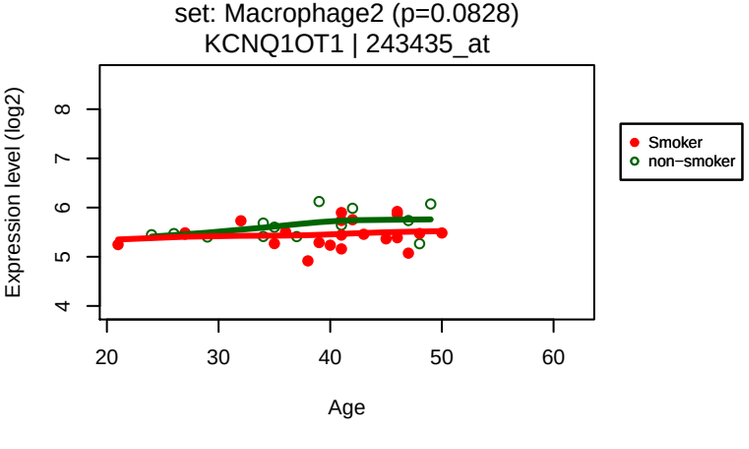

set: SmallAirway2 (p=0.8202)
KCNQ1OT1 | 243435_at
Expression level (log2)
Age
Smoker
non−smoker

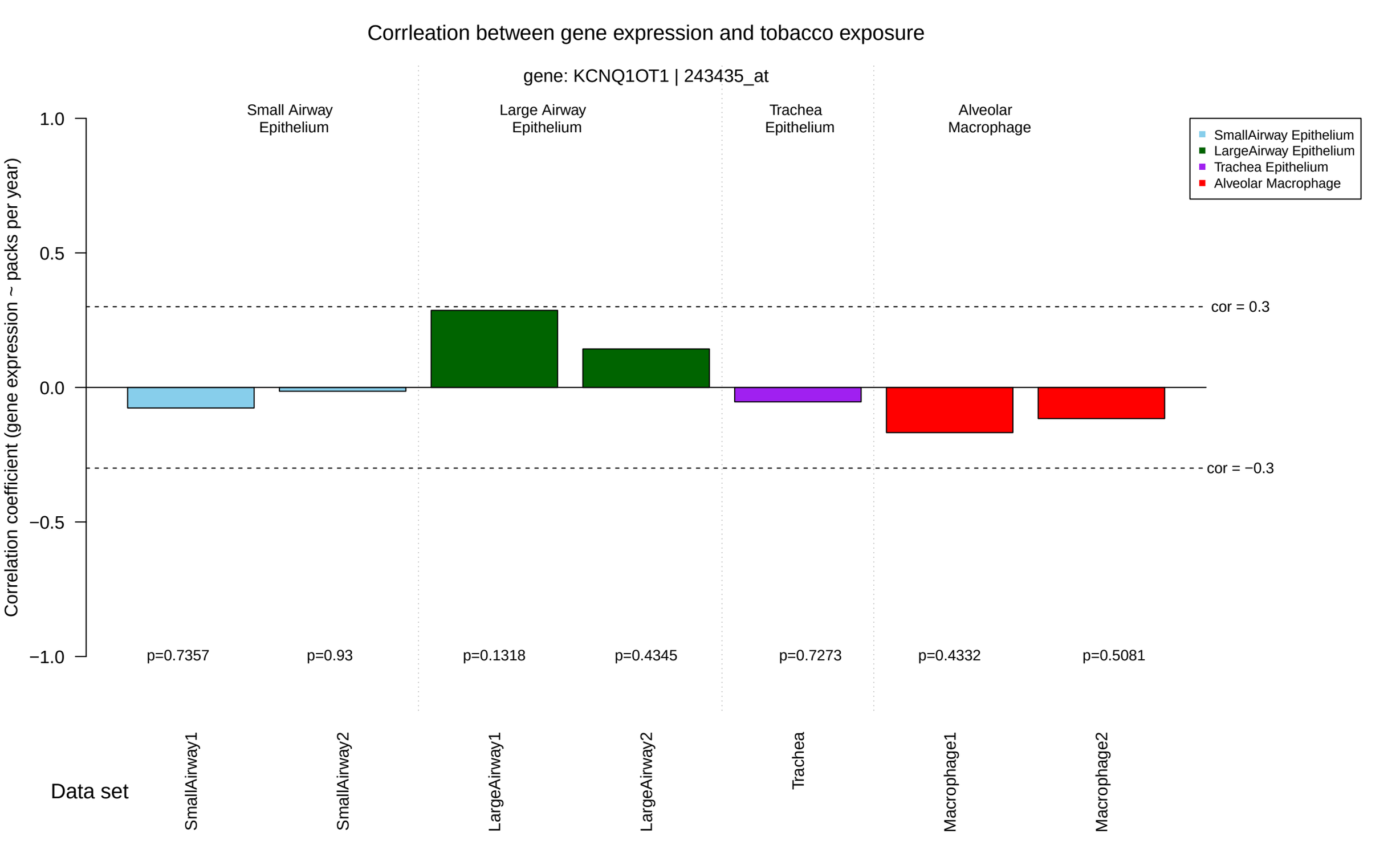

set: LargeAirway1 (p=0.1363)
KCNQ1OT1 | 243435_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9365)
KCNQ1OT1 | 243435_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4146)
KCNQ1OT1 | 243435_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1532)
KCNQ1OT1 | 243435_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0828)
KCNQ1OT1 | 243435_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNQ1OT1 | 243435_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7357
p=0.93
p=0.1318
p=0.4345
p=0.7273
p=0.4332
p=0.5081
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2